

Microbial Diversity in Vineyard Soil

Team Better Wine

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Abstract—Sustainable farming practices are essential in minimizing the negative environmental impacts of farming and food production, and guarantee long term welfare of the ecosystem. Soil health, which is supported by soil microorganisms, is essential in developing sustainable farming practices.

Our findings below focus specifically on the characterization of soil microbial communities in a central California vineyard and the influence of varying viticulture practices. Our analysis is driven by the identification of organisms that drive differences in the variability in community composition among samples in a habitat (beta-diversity).

The foundation of our analysis lies in our calculation of differential correlations, which provided a way to enumerate microorganisms that saw similar changes in correlation between experiments. We analyzed microorganism abundance correlations across differing experiments on various taxonomic levels and used hierarchical clustering to yield groups, or “modules” of microorganisms with significant correlations. We accessed the Functional Annotation of Prokaryotic Taxa (FAPROTAX) database to predict microorganism functionality and assess the significance of our constructed modules. We examined edaphic quantities such as rainfall and carbon-nitrogen levels in relation to patterns in differential correlation. We aimed to answer questions regarding beta-diversity from the construction of our modules and edaphic quantities, including how microorganisms interact within common groups and additional external factors that may have driven variability in community composition across treatments.

I. INTRODUCTION

Sustainable farming practices are crucial in minimizing the impact of agriculture production and processes on the wellbeing of our environment. A key factor in sustainable farming practices is understanding soil health, which is essential for crop productivity and longevity. Soil health is largely influenced by the variety and abundance of different microorganisms in the soil. These microorganisms perform essential functions within the soil, such as providing nutrients, decomposing matter and preventing disease.

Our study of interest performed soil sampling at a local vineyard, measuring microbial community

composition timed with the grapevine life cycle across three years. Three different soil experiments were conducted across the sampling vineyard, testing the effect of herbicide presence, crop cover, and type of fertilizer on the abundance of microorganisms in the soil. Three soil samples were collected per treatment during each of the four stages of the vineyard cycle: bud break, bloom, veraison and harvest. In total, the study included 96 soil samples, each of which were analyzed by a biochemistry team to mark the abundance of each microorganism found in each sample.

Soil microbial diversity is generally measured by alpha and beta diversity. Alpha-diversity refers to the observed number or relative abundances of taxa of an average sample within a habitat type. Beta-diversity measures differences in microbial community composition between samples. The measure of both alpha and beta diversity would provide a cohesive and complete picture of soil microbial diversity, allowing farmers to better understand how to improve their practices.

Previous external analysis of this data used permutation tests to determine if the number of taxa or relative abundance of each taxa, referred to as alpha-diversity, changed across treatments for the three experiments. The results of the permutation testing yielded no statistically significant evidence of difference in alpha-diversity across treatments, suggesting that the overall richness and evenness of microorganisms remained relatively constant regardless of treatment. However, beta-diversity, which measures differences in microbial community composition between samples, was not explored in the original analysis. Beta-diversity provides a different measure of soil microbial diversity than previously investigated, motivating our assessment in identifying drivers of differences in microbial community structure across samples and treatments.

Our objective was to determine whether beta-diversity varied across treatments and to assess whether specific farming practices led to different microbial community structures and microorganism interactions.

II. DATA DESCRIPTION

The provided data was derived from one primary dataset. Each observation represents an ordered taxonomic unit (OTU) and the relative abundance of a specific microorganism for a single sample. Each entry contains key information such as the taxonomic identity of each microorganism, the treatment type for the given sample, the year the sample was collected (between 2016 and 2018), the sample replicate, the stage in the harvesting cycle at the time of collection, and the number of microorganisms of the specified class in the sample.

Due to the large number of OTUs present in each soil sample, the original long-formatted dataset was difficult to manage and perform analysis on. To solve this, the data was modified for usability with rarification. This process involved retaining only the top 50 OTUs in each sample, rather than counting every microorganism that appeared. Rarified counts were provided for each sample under different conditions, with separate datasets grouped by individual taxa at the Class, Order, Family, and Genus levels. For our analysis, we primarily used the rarified data grouped by taxonomic class, which is a refined subset of the primary dataset containing information about the observed count of microorganisms belonging to each class.

III. DIFFERENTIAL CORRELATION

Measuring the overall variability in community composition was the first step in identifying drivers of differences in beta-diversity. Differential correlation, or the difference between correlations under two treatments, measures the change in pairwise correlation between two microorganisms under two different conditions. To compute differential correlation, we calculated the cell-wise difference between two correlation matrices. High magnitude differential correlation between two classes suggests that under the two conditions, the relationship shifts dramatically. This shift suggests the way the microorganisms interact with each other changes depending on the treatment it receives.

To begin our analysis, we assessed microorganism abundance correlations as a way to confirm trends across samples to explain beta-diversity did in fact exist. To simplify our computations and analysis, we honed in on the herbicide experiment specifically for the majority of our analysis. The herbicide experiment has two treatments (herbicide/non-herbicide) whereas other experiments had three, and avoided extensive pairwise comparisons across three treatments. We computed all pairwise correlations in abundance between the 78 microorganism classes under the herbicide and non-herbicide treatments. The result was two 78x78 correlation matrices, each cell representing

the correlation between two microorganisms under the respective treatment. Two microorganisms simultaneously appearing more or less indicated that the two microorganisms were correlated under the respective experiment. Figure 1 and Figure 2 display subsets of the 78x78 correlation matrix for readability. The lower right regions of the two matrices can be compared, and indicate that the abundances of those microorganisms were simultaneously higher in the non-herbicide treatment versus the herbicide treatment.

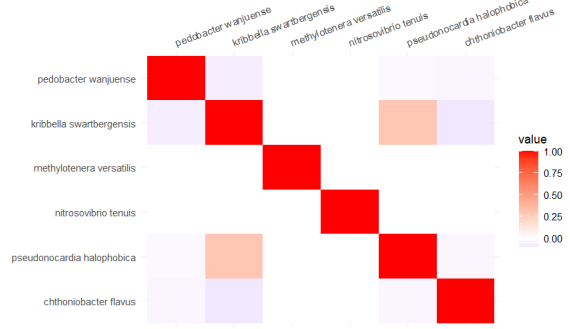


Fig. 1. Herbicide Correlation Matrix

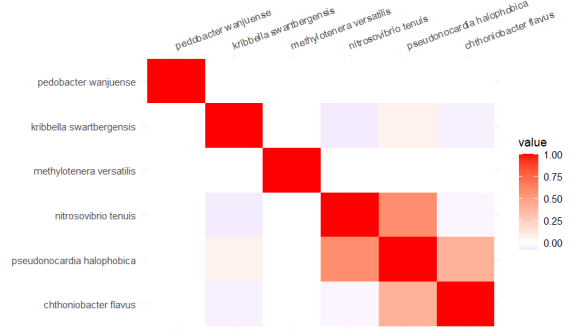


Fig. 2. Non-Herbicide Correlation Matrix

Figure 3 displays a subset of the resulting differential correlation matrix for the Herbicide/Non-Herbicide treatment for readability. The complete differential correlation matrix for the Herbicide/Non-Herbicide treatment can be referenced in the appendix alongside complete differential correlation matrices for all pairwise treatments for all experiments in this study. Note here, a high in magnitude positive differential correlation indicates a positive correlation between two microorganisms in the herbicide treatment and a negligible correlation in the non-herbicide treatment. A high in magnitude negative differential correlation indicates a negligible correlation between two microorganisms in the herbicide treatment and a positive correlation in the non-herbicide treatment. As an example, the differential correlation between acidimicrobiia

and armatimonadia is 1.7. We can conclude that the correlation between acidimicrobii and armatimonadia differed across treatments. Acidimicrobiia and armatimonadia had a positive correlation in the H treatment, and negligible correlation in the NH treatment.

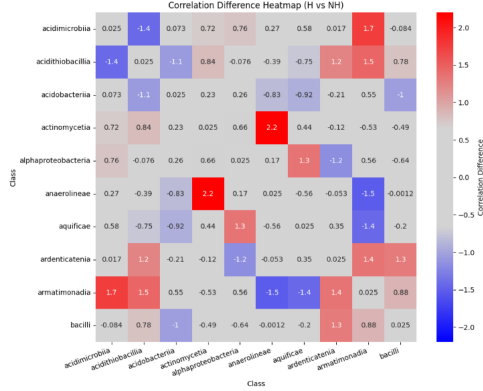


Fig. 3. Differential Correlation Matrix

A. Modules

After computing differential correlations under the herbicide versus non-herbicide treatments, our goal was to group microorganisms that saw similar changes in correlation between two treatments. We computed the correlation z-scores for each cell in the difference matrix, using formula (1), and leveraged the normal distribution to obtain p-values for each cell. The resulting p-value matrix quantifies the statistical significance of each pairwise differential correlation between microorganisms across two treatments. Each entry in this matrix represents how significantly different the correlation is for a given pair of microorganisms between treatments.

$$\frac{r_1 - r_2}{\sqrt{1 - r_1^2} + \sqrt{1 - r_2^2}} \quad (1)$$

Instead of using traditional distance metrics for our clustering algorithm, we treated these p-values as our metric, measuring how similarly two microorganisms of the same taxa responded to treatment exposure. Lower p-values indicate stronger evidence that correlation patterns changed similarly across conditions, implying a shared response between the two classes across treatments. This is visualized by the plot in figure 4, showing abundance over time for group 10. This plot demonstrates clear negative correlation (non-simultaneous spikes) under the herbicide treatment and strong positive correlation (simultaneous spikes) under the non-herbicide treatment, indicating high differential correlation. This gave us confidence that the modules we constructed were detecting the differences in beta-diversity that we were targeting, and that our module building procedure was operating well.

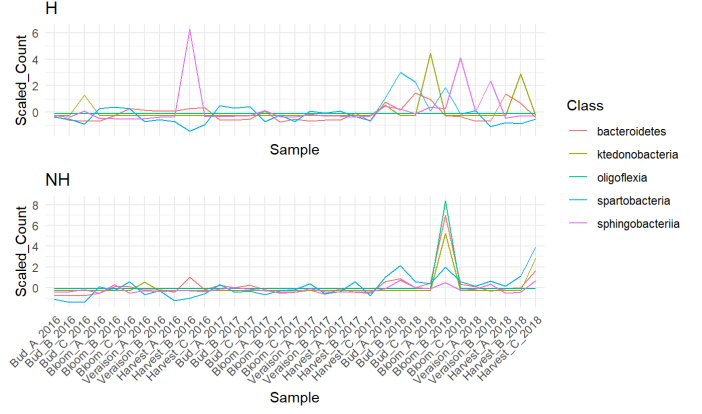


Fig. 4. Abundance Over Time Plot

Instead of using traditional distance metrics for our clustering algorithm, we treated these p-values as our metric, measuring how similarly two microbial classes responded to herbicide exposure. Lower p-values indicate stronger evidence that correlation patterns changed similarly across conditions, implying a shared response between the two classes across treatments.

We used the p-value distance matrix to perform hierarchical clustering on the classes, merging classes with the smallest p-value distance to build the clusters and hierarchical structure. Figure 5 depicts a branch of the final dendrogram containing module 17 and module 10 respectively. The dotted red line is indicative of our p-value threshold of 0.1. The complete hierarchical clustering dendrogram can be referenced in the appendix.

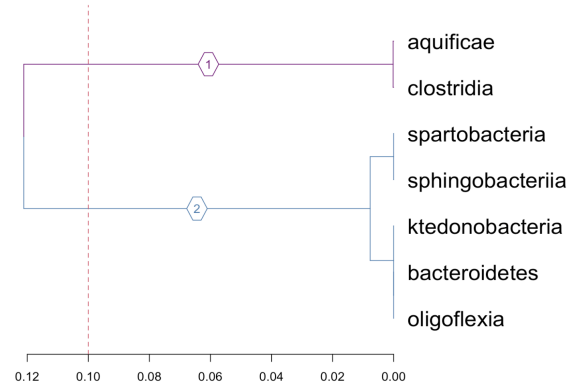


Fig. 5. Hierarchical Clustering of Modules 17 and 10

After honing our procedure using the more simple herbicide/non-herbicide experiment, we extended our analysis to new experiments and taxa levels. The module formation procedure was expanded to cover all treatment types: herbicide, fertilizer, and crop cover. In total, we produced modules across 14 distinct pairwise treatment conditions at the class level. Modules at the class

level provide useful information and allow us to observe trends in functionality, correlation, and the ways in which treatments affect microbial interactions; however, we also sought to identify order level modules that could provide more granular information about microbial communities across the pairwise treatment combinations. Of note, in general, there were more modules identified at the order level compared to the class level. This finding is consistent with the fact that there is more microbial diversity at the order level and applying our filtering procedure identifies which of these taxa are most interactive and responsive to manipulations in experimental conditions.

B. Module Functionality

The construction of modules provided a concrete measure of variability in community composition among samples and a statistical case for our analysis, however, drivers of differences in beta-diversity still had not been identified. We expand our module analysis in addressing microbial functionality within our previously constructed modules. Functional profiles of microbial communities are extremely helpful in explaining the structure and dynamics of microbial communities and address our objective of identifying drivers of beta-diversity.

The Functional Annotation of Prokaryotic Taxa (FAPROTAX) database is a collection of established metabolic or other ecologically relevant functions, using the current research and literature on cultured strains, derived from 16s rRNA amplicon sequencing. Choosing to work with the FAPROTAX database was motivated by the complicated and expensive nature of metagenomic sequencing, while amplicon sequencing sequences the 16s rRNA gene across multiple species and is a promising alternative [1]. We mapped the taxonomy of the microorganisms within each module against the FAPROTAX database to predict the traits and functions on biogeochemical roles in the soil. The FAPROTAX database yielded microorganism functionality on various taxonomic levels, although was the most informative on the genus level.

The construction of our modules on the class level proved to be an issue as we compared them against the FAPROTAX database. The available research and literature largely regarded functionality on the species level, causing our initial mapping to be especially sparse. However, we used the classes defined in our modules to climb down the taxonomic hierarchy to instead map the OTUs against the FAPROTAX database on the genus level. From here, we were able to conclude the functionality of microorganisms in our modules on the class level based on the OTU's genus.

The modules outlined in Figure X exhibit high statistically significant differential correla-

tions. This indicates that the microorganisms in each of those modules tended to appear together simultaneously under the herbicide treatment but appeared incongruously in non-herbicide soil treatment samples. Figure X depicts the five microorganisms in Module 10 and their respective functionalities. Functional profiles of all 28 modules created on the class level can be referenced in the appendix. Overlapping functionalities confirmed our previous module construction to be meaningful and provided insight regarding the microbial communities within our modules. Shared functional profiles are helpful for understanding which microorganisms drive differences in community composition between samples (beta-diversity).

Module 10 Functionality	
Microorganism	Traits and Functions
sphingobacteria	aerobic chemo-heterotrophy, nitrate reduction, ureolysis, xylanolysis, cellulolysis
spartobacteria	aerobic chemoheterotrophy
oligoflexia	nitrate reduction, predatory or exoparasitic
Bacteroidetes	aerobic chemo-heterotrophy, xylanolysis, cellulolysis
ktdonobacteria	aerobic chemo-heterotrophy, nitrate reduction

Fig. 6. Functional Profile of Module 10

We chose to define a similarity score to measure overlapping functionalities across our modules. We define the similarity score to be the number of shared functionalities within a module out of all functions found, or the proportion of shared functions. Figure X displays respective modules with its corresponding functionality similarity score, for 12 out of 28 modules. The complete table of modules and similarity scores can be referenced in the appendix. We can see here that the proportion of microorganisms in Module 10 that had overlapping functionalities is 0.66.

While the similarity score provided a coherent measure of module similarity, it was rather elementary. The similarity score measured functions shared between just two microorganisms and a function shared between all microorganisms within

a module the same way. Distinguishing modules with shared functions across many microorganisms would further our identification of drivers of beta-diversity. We created a similarity metric as a way to measure shared functionalities in a way that prioritized modules with functionalities shared among many microorganisms and penalized microorganisms with functionalities not shared with any other microorganisms. The similarity metric is indicative of the percent of function instances that are shared.

To calculate this metric, we visualized our modules as a graph where microorganisms are nodes and edges connecting them represent a shared functionality between two microorganisms. A self-edge in this representation is indicative of a microorganism having a unique function not shared by any other microorganism within the respective module. We calculated the quotient of connected edges and both connected and unconnected edges to represent functionalities belonging to more than one microorganism. To compute the similarity metric, modules were represented in matrix form.

Let A denote the matrix representation of a module. A module with i microorganisms and j functions in the complete functional profile (all functions found in a module, regardless of microorganism responsible for the function) will be of dimension $(i \times j)$. The ij th entry is denoted 1 if microorganism i is responsible for function ij , and denoted 0 if microorganism i is not responsible for function j . The cross product of A and A^T was taken, denoted $B = Ax A^T$. Note, matrix B has dimension $(i \times i)$ where the nm th entry of matrix B indicates the number of functionalities microorganism n shares with microorganism m where $n, m \in \{1, \dots, i\}$. Finally, the similarity metric is calculated using equation 2.

$$\frac{\sum_{i=1}^n \sum_{j=i+1}^n B_{ij}}{\sum_{i=1}^n B_{ii}} \quad (2)$$

To provide a statistical case for the similarity metric, random sampling was performed to test the significance of the construction of our modules. Random samples of the same size as each module were created to calculate the proportion of random samples that had a similarity metric at least as extreme as the similarity metric observed for our constructed modules (standard p-value calculation). Figure 7 displays each module's respective functionality similarity metric and respective p-value, for 12 out of 28 modules. The complete table of modules and their similarity metric and p-values can be referenced in the appendix. We can see the percent of function instances in Module 10 that are shared is 45.84%, yielding a p-value of 0.0033.

Module	Similarity Score	P-Value
10	0.4583	0.003
3	0.2545	0.060
11	0.2432	0.080
12	0.2222	0.027
27	0.1627	0.097
13	0.1538	0.290
7	0.1333	0.130
5	0.1250	0.397
17	0.1200	0.157
2	0.1190	0.163
9	0.1097	0.463
4	0.0983	0.540

Fig. 7. Similarity Scores and P-Values by Module

C. Differential Correlation Investigation

We noticed that in almost all cases where differential correlation was high, microbial classes tended to move together under one treatment and separately under the other in early 2018, particularly during the bud and bloom seasons. This is shown clearly in previous figure X (abundance over time plot). To explore potential drivers of this trend in differential correlation, we focused on two environmental factors: rainfall data from the area surrounding the vineyard and carbon-nitrogen (C-N) ratios collected directly from the soil during sampling. To test whether these factors might be contributing to the observed patterns, we calculated the contribution to correlation over time for each microbial class within modules under different treatment conditions. This contribution metric quantifies how much a specific sampling period in (ex. Bud 2017) adds to the overall differential correlation between treatments. Since the replicates weren't aligned between treatments (e.g., Replicate A under herbicide had no relation to Replicate A under non-herbicide), we averaged the correlation contribution across all three replicates for each time point. The contribution to correlation formula in equation 3, with u_i representing the correlation between the two taxa under treatment 1 during sampling period i and v_i representing the correlation under treatment 2.

$$r = \frac{1}{n} \sum (u_i v_i) \quad (3)$$

Resulting sample periods of interest would have high differential correlation at that specific period, or in other words, a large difference between u_i and v_i , signifying a large contribution to differential correlation at this time period. We then

overlaid these seasonal contribution values with both rainfall and C-N data to look for temporal alignment—essentially, seeing if environmental fluctuations lined up with periods of high difference between contribution to correlation under treatment 1 and treatment 2. Figure X is an example of the contribution to correlation overlaid with the yearly cumulative rainfall for module 6.

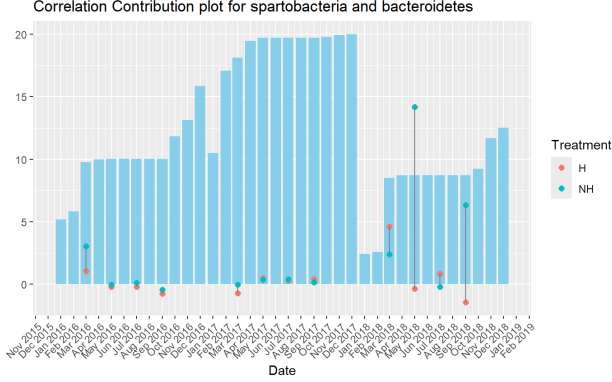


Fig. 8. Correlation Contribution for spartobacteria and bacteroidetes

We analyzed these correlation contribution plots overlaid with edaphic data for each module under the herbicide experiment at the microbial class level. While some trends appeared visually, we ultimately didn't find any evidence that rainfall or carbon-nitrogen levels were consistently driving the differences in correlation across treatments. As we stated earlier, we saw a trend of high contribution to correlation in 2018 across almost all pairwise class comparisons within each module. However, rainfall had actually peaked in 2017 and C-N didn't have any clear peaks, so the timing of anomalies in edaphic data didn't align with the periods of high correlation shifts. This disconnect suggests that other factors, such as differences in herbicide application, unmeasured climatic variables, or even changes in soil chemistry, may be responsible for the observed microbial diversity shifts. While rainfall and C-N are important edaphic factors, our analysis suggests that neither individually seemed to explain the variation in microbial beta-diversity at different sample periods. Future work could explore additional environmental variables like temperature, pH, herbicide composition, or microbial functional traits to potentially find the true drivers of differential correlation in these soil communities.

IV. CONCLUSION

Minimizing the negative impact of farming and food production is crucial for the long-term welfare of the ecosystem. Understanding soil health plays a crucial role in sustainable farming practices and is essential to both farmers and researchers. The variety and abundance of microorganisms in soil

largely dictate soil health as these microorganisms are responsible for functions that make soil fertile, resilient, and capable of plant growth. With microbial abundance data from a Central California vineyard, we were motivated to identify drivers of differences in microbial community structure across samples and treatments (beta-diversity). We began our identification of drivers of beta-diversity with the definition and calculation of differential correlation in this context. Differential correlations measured the pairwise changes in correlation of microorganisms across treatments, and motivated a large majority of our analysis. These differential correlations motivated our use of hierarchical clustering to group together microorganisms that saw similar changes in correlation between two treatments, or modules of microorganisms. Mapping our modules against the FAPROTAX database to predict microbial functionality provided insight into how these microorganisms were interacting with one another and the variability in community composition. High differential correlations, driven by simultaneous spikes or deviances in abundance may be linked to an increased abundance of certain microorganisms caused by differences in climate, weather and other uncontrolled environmental variables. Edaphic qualities such as rainfall and carbon-nitrogen levels in the soil were then examined alongside patterns of differential correlation to identify external factors that may have driven variability in community composition across treatments.